

27 Ans: A

$$E = -\frac{d\phi}{dt} = -\frac{0 - BA}{10\text{ ms}} = \frac{(5)(\pi \times 0.08^2)}{10 \times 10^{-3}} = 10.05\text{ V}$$

$$\text{Induced current, } I = \frac{E}{R} = \frac{10.05}{4} = 2.51\text{ A}$$

Clockwise direction, apply Lenz's Law. The flux is getting smaller, so the induced current will try to increase the flux by generating a magnetic field that is in the same direction as the external field.

28 Ans: B